Partial ordered sets (POSETS) – : A non empty set X with a relation R is said to be partially ordered set or POSET if following properties are satisfied

(1) Reflexivity – : $aRa \ \forall a \in X$

(2) Antisymmetry – : If for some $a, b \in X$, aRb & $bRa \Rightarrow a = b$

(3) Transitivity – : If $a, b, c \in X$ s.t. $aRb, bRc \Rightarrow aRc$

Q – Let R be a relation "divides" in the set $X = \{2, 3, 4, 6, 12, 24, 36\}$. Show that (X, R) is a poset.

Sol In order to prove (X, R) is a poset, we shall prove that R is a partial order relation on X

(i) Reflexivity – If $a \in X$, then 'a' divides 'a'

$\Rightarrow aRa \ \forall a \in X \quad \therefore R$ is reflexive

(ii) Antisymmetry – If $a, b \in X$ & aRb & bRa

$\Rightarrow$ a divides b & b divides a

$\Rightarrow$ we get positive Integers k_1, k_2 s.t

$$\frac{b}{a} = k_1 \text{ — ①} \quad \& \quad \frac{a}{b} = k_2 \text{ — ②}$$

multiplying ① & ②

$$\frac{\cancel{b}}{\cancel{a}} \times \frac{\cancel{a}}{\cancel{b}} = k_1 k_2$$

$\therefore k_1 k_2 = 1 \Rightarrow k_1 = k_2 = 1$ definitely

& when $k_1 = k_2 = 1$

from ① & ② both, we have $a = b$

$\therefore R$ is antisymmetric

(iii) Transitivity – If $a, b, c \in X$ & aRb & bRc

$\Rightarrow$ a divided b & b divides c

$$\Rightarrow \frac{b}{a} = k_1 \text{ — ①} \quad \& \quad \frac{c}{b} = k_2 \text{ — ②}$$

multiplying ① & ②

$$\frac{\cancel{b}}{a} \times \frac{c}{\cancel{b}} = k_1 k_2 \Rightarrow \frac{c}{a} = k_1 k_2$$

$\Rightarrow$ a divides c $\Rightarrow aRc \Rightarrow R$ is transitive

$\therefore R$ is reflexive, antisymmetric & transitive on X

$\therefore (X, R)$ is a poset by above definition of poset

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Q Show that relation

$$R = \{(1,1), (2,2), (3,3), (4,4), (3,1), (1,4), (3,2), (3,4)\} \text{ in } X = \{1,2,3,4\}$$

forms a poset (X, R)

Sol – (i) Reflexive – : $\because (1,1), (2,2), (3,3)$ & $(4,4) \in R$ $\therefore$ R is reflexive

(ii) Antisymmetric – : If $(a,b) \in R$ & $(b,a) \in R$, then $a = b$

(iii) Transitive – : $(3,1) \in R$, $(1,4) \in R \Rightarrow (3,4) \in R$

similarly $(1,1) \in R$, $(1,4) \in R \Rightarrow (1,4) \in R$ etc.

$\therefore$ R is transitive also

~~when~~ Thus R is a partial order relation on X $\therefore (X, R)$ is a poset

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Comparable – Two elements a & b of a poset $(X, \leq)$ are said to be comparable if either $a \leq b$ or $b \leq a$

for example In poset $(Z^+, |)$, the integers 3 & 9 are comparable

$\because$ here relation is '|' i.e divides & 3 divides 9, so 3 & 9 are comparable

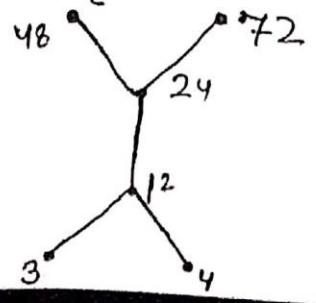
but 5 & 7 are non comparable because neither 7 divides 5 nor 5 divided 7.

Hasse Diagram – : (Pictorial representation of a Poset)

A graphical representation of a poset $(X, \leq)$ is called Hasse diagram. In such diagrams we represent ~~each~~ each element of X by a dot (•) in such a way that if $a \leq b$ then dot of a lies below b & if b divides a then 'b' should below 'a'. & join these dots by straight lines. Dots of incomparable elements are not joined by straight lines. Also we do not join edges which imply transitivity.

Q – Draw the Hasse diagram of $(A, \leq)$ where $A = \{3, 4, 12, 24, 48, 72\}$ & relation $\leq$ is such that if $a \leq b$ then a divides b

Sol – First element of A is 3 so take 3 at bottom, then we have 4 & 3 does not divided 4 so we put 4 separately not above 3 again 3 & 4 both divides 12 $\therefore$ 12 is placed in middle of 3 & 4 & above 3 & 4 & then we have 24, 12 divides 24, & then there is 48 & 72, 24 divides both



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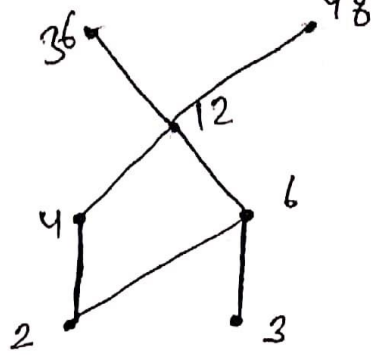
Q - Draw Hasse diagram of relation S defined as "divides" on

Set $B = \{2, 3, 4, 6, 12, 36, 48\}$



Sol. $\because$ 2 does not divide 3

So 2 & 3 are placed separately

then we have 4 & 2 divides 4

So 4 is placed above 2

& 3 divides 6 so 6 is placed above 3

also 2 divided 6 $\therefore$ we join 2 & 6 also

now we have 12, both 4 & 6 divided 12.

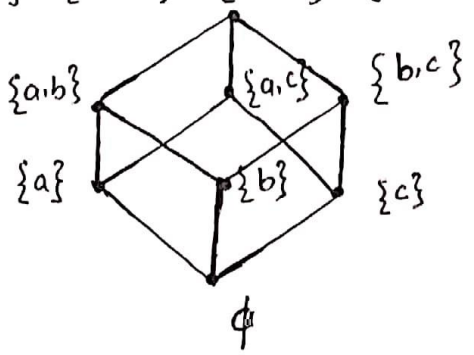
$\therefore$ 12 is placed above 4 & 6 in middle, again, we have 36 & 48

both are divided by 12

Q - Draw Hasse diagram of the poset $\{P(X), \subseteq\}$, where $X = \{a, b, c\}$

Sol - $P(X)$ = power set of X i.e all subsets of X

$= \{\phi, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a, b, c\}\}$



Now there are some definitions

(i) Minimal & maximal element of a poset - : An element a in a poset $(X, \leq)$ is called a minimal element if there is no element c in X s.t $c \leq a$

ex - in poset $\{2, 3, 4, 6, 12, 24, 36, 48\}$ with divisibility relation 2, 3 are minimal elements because 4, 6, 12, 24, 36, 48 are divisible by either 2 or 3 $\therefore$ both 2, 3 are minimal elements

if 3, 4, 6, 12, 24, 36, 48 are divisible by 2 then 2 will be minimal element but 3 is not divisible by 2 so 2 is not only minimal element

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